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Student-produced response directions

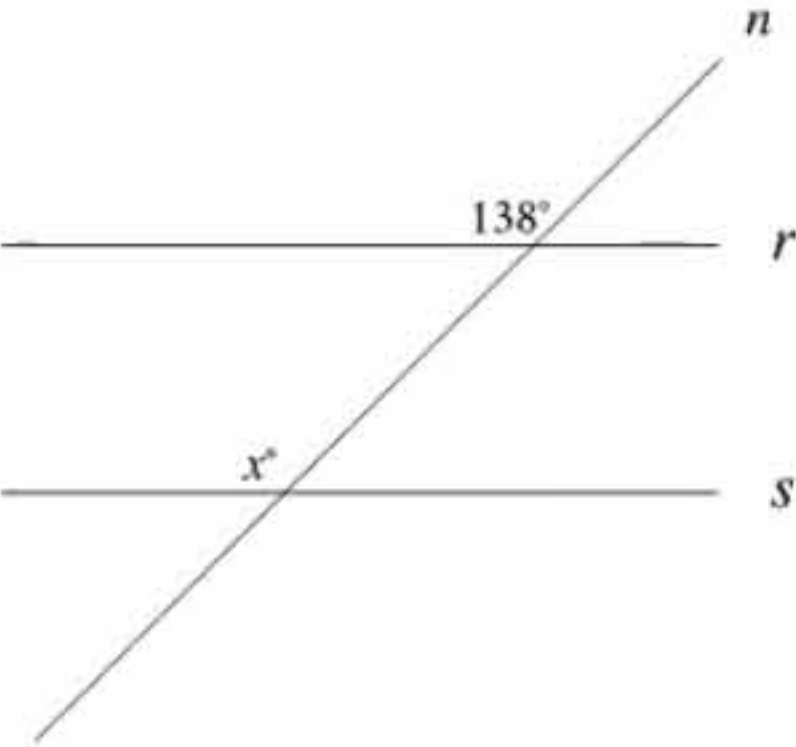
- If you find **more than one correct answer**, enter only one answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 ~ ~ ~ ~	0.66 .66 0.67

1

Mark for Review



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

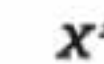
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Mark for Review



During a portion of a flight, a small airplane's cruising speed varied between 125 miles per hour and 135 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

(A) $s \leq 10$

(B) $s \leq 125$

(C) $s \leq 135$

(D) $125 \leq s \leq 135$



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Mark for Review



If $5x = 4$, what is the value of $40x$?

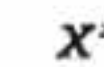
- A. 4
- B. 12
- C. 28
- D. 32



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Mark for Review



Which expression is equivalent to $(x^3 + 4x^2 - 3x) + 5(x^2 + 8)$?

(A) $x^3 + 9x^2 - 3x + 40$

(B) $x^3 + 4x^2 - 3x + 40$

(C) $x^3 + 8x^2 - 3x + 40$

(D) $x^3 + 5x^2 - 3x + 40$



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5

Mark for Review

The function f is defined by $f(x) = x^3 + 14$. What is the value of $f(2)$?

Answer Preview:



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Mark for Review

The function g is defined by $g(x) = \frac{x}{2}$. For what value of x does $g(x) = 624$?

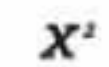
Answer Preview:



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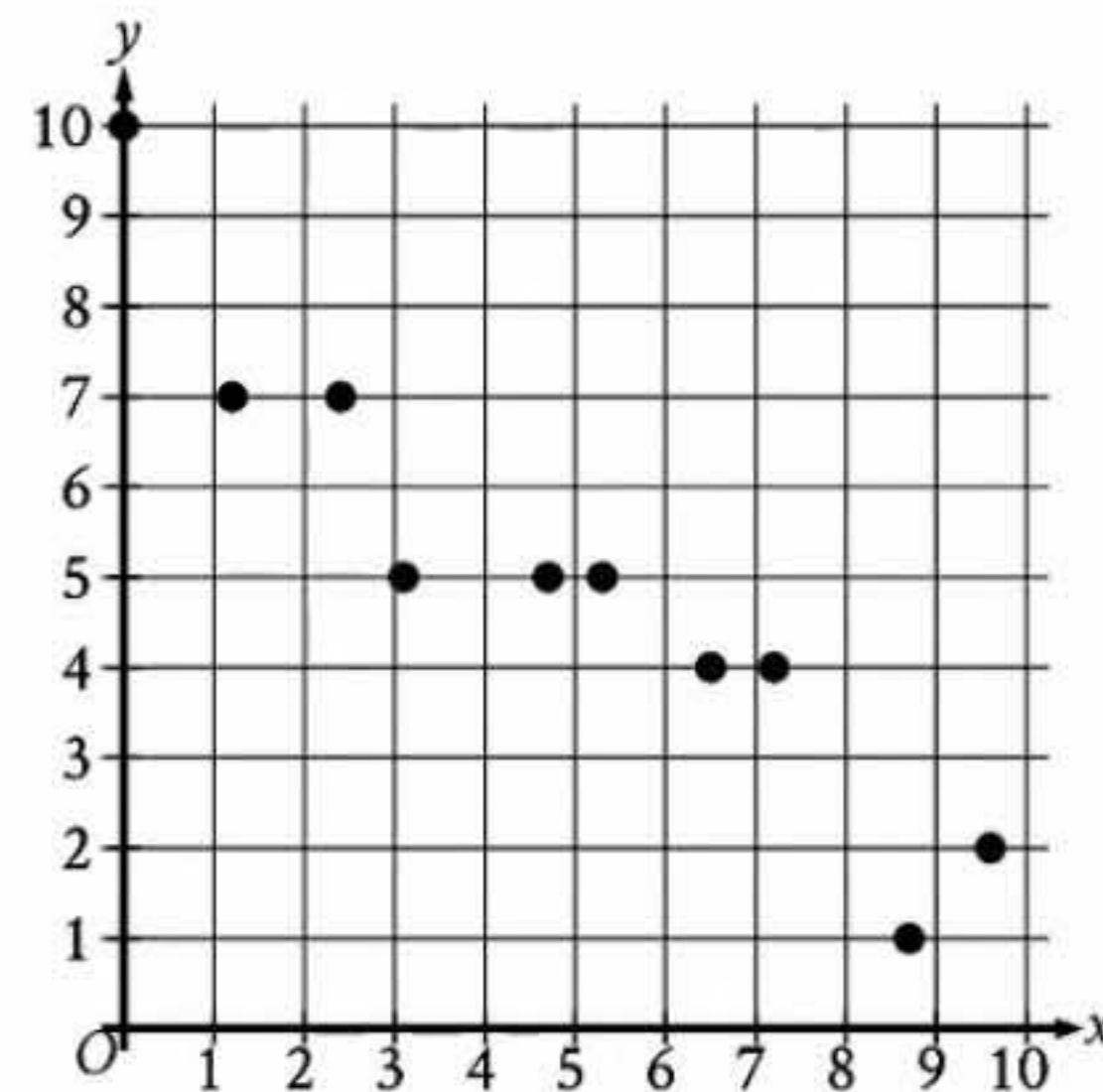


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7. The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

A. $y = 0.8 + 8.7x$

B. $y = 0.8 - 8.7x$

C. $y = 8.7 + 0.8x$

D. $y = 8.7 - 0.8x$



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Mark for Review



The function $f(x) = \frac{1}{9}(x - 4)^2 + 5$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- (A) The toy car's minimum height was 5 inches above the ground.
- (B) The toy car's minimum height was 4 inches above the ground.
- (C) The toy car's height was 5 inches above the ground when it started moving.
- (D) The toy car's height was 4 inches above the ground when it started moving.



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Mark for Review

$7x - 5y = -70$

For the given equation, which table gives three values of x and their corresponding values of y ?

- A

x	0	5	10
y	14	21	28
- B

x	0	5	10
y	28	21	14
- C

x	14	21	28
y	0	5	10
- D

x	14	21	28
y	10	5	0



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10

Mark for Review

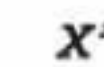
A chemist combines water and isopropanol to make a mixture with a volume of 45 milliliters (mL). The volume of isopropanol in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)



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Mark for Review

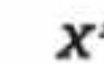


$$y = -\frac{1}{7}x$$

$$y = \frac{1}{11}x$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -7
- B. 0
- C. 4
- D. 11



12



Mark for Review



A company has a customer loyalty program. In January 2018, there were 1,600 customers enrolled in the loyalty program. For the next 24 months after January 2018, the total number of customers enrolled in the loyalty program each month was 5% greater than the total number enrolled the previous month. Which equation gives the total number of customers, c , enrolled in the company's loyalty program m months after January 2018, where $m \leq 24$?

(A) $c = 1,600(0.05)^m$

(B) $c = 1,600(1.05)^m$

(C) $c = 1,600(1.5)^m$

(D) $c = 1,600(5)^m$



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13

Mark for Review

The function $f(t) = 10,000(2)^{\frac{t}{850}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

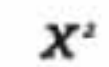
Answer Preview:



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Mark for Review



$$y = \frac{5}{7}x + 9$$

One of two equations in a system of linear equations is given. The system has infinitely many solutions. What is the slope of the graph of the second equation?

(A) $-\frac{7}{5}$

(B) $-\frac{5}{7}$

(C) $\frac{5}{7}$

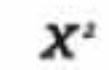
(D) $\frac{7}{5}$



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Mark for Review



If $4x^2 - 12x - 24 = 0$, what is the value of $x^2 - 3x$?

(A) 0

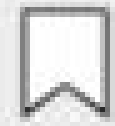
(B) 3

(C) 4

(D) 6

16.

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Mark for Review

The length of each edge of a box is 29 inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?



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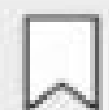
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Mark for Review

Raheem bought 9 shirts that were each the same price. He used a coupon for \$54 off the entire purchase. The cost for the entire purchase after using the coupon was \$108. What was the original price, in dollars, for 1 shirt?

18.

18



Mark for Review



What is the value of $\tan \frac{92\pi}{3}$?

(A) $-\sqrt{3}$

(B) $-\frac{\sqrt{3}}{3}$

(C) $\frac{\sqrt{3}}{3}$

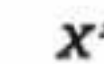
(D) $\sqrt{3}$



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Mark for Review



In right triangle ABC , angles A and B are acute, side AC has a length of 21.2, and $\tan B = \frac{1}{6}$. What is the length of side BC , rounded to the nearest tenth?

20.

20



Mark for Review



$$(x + 4)^2 + (y - 19)^2 = 121$$

The graph of the given equation is a circle in the xy -plane. The point (a, b) lies on the circle. Which of the following is a possible value for a ?

(A) -16

(B) -14

(C) 11

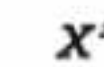
(D) 19



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Mark for Review



$$f(x) = (x - 4)^2 + 11$$

What is the minimum value of the given function?

(A) 4

(B) 7

(C) 11

(D) 15



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Mark for Review



Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

(A) $\frac{25\pi}{2}$

(B) $\frac{27\pi}{2}$

(C) 22π

(D) 23π